Rigorous Convergence Analysis of Federated Learning with Partial Forgetting Compensation (FL-PFC)

***Abstract*—This paper presents a rigorous convergence analysis for federated learning with partial forgetting compensation (FL-PFC) under three challenges: Non-IID clients, partial participation, and proxy-based knowledge distillation. The analysis proceeds in two steps: first establishing convergence on a fixed surrogate objective, then extending to the actual adaptive objective via a conditional-expectation-freezing technique. Without relying on the decay of dynamic weight bias, only a constant bias upper bound is used, yielding the conclusion that the algorithm converges to an $\mathcal{O}(G^2)$ error neighborhood rather than an exact stationary point. The main body presents core theory with proof sketches, while complete step-by-step derivations appear in the Appendix.**

I. Convergence Analysis

Problem formulation. Consider K clients. In communication round t, a subset S_t of clients is sampled for training. Client k minimizes the local objective

$$\widetilde{\mathcal{L}}_t^k(\omega;\omega_t) = \mathcal{L}_C^k(\omega) + \lambda_t^k \mathcal{L}_{\mathrm{KD}}^k(\omega) + \frac{\mu_t}{2}\|\omega-\omega_t\|^2, \tag{1}$$

where $\mathcal{L}_C^k$ denotes the classification loss, $\mathcal{L}_{\mathrm{KD}}^k$ the knowledge distillation loss, and the last term a proximal regularizer. Define two analytical objects: the adaptive round-wise objective $\widetilde{F}_t(\omega) \triangleq \sum_k p_k\,\mathbb{E}_\xi[\widetilde{\mathcal{L}}_t^k]$ and the time-uniform reference objective $F(\omega) \triangleq \sum_k p_k F_k(\omega)$ with $p_k = n_k/n$.

Let $g_{t,e}^k$ denote the stochastic gradient, $\bar{g}_t^k \triangleq \frac{1}{E}\sum_{e=0}^{E-1} g_{t,e}^k$, and $g_t \triangleq \sum_k p_k \bar{g}_t^k$ the ideal full-participation gradient. Define the actual aggregated gradient $d_t \triangleq \sum_k \alpha_{k,t}\bar{g}_t^k$, bias $b_t \triangleq d_t - g_t$, and proxy residual $r_t \triangleq \beta_t(\omega_t^{\mathrm{proxy}} - \omega_t)$. Weights $\alpha_{k,t}, \beta_t$ follow dynamic weighting and the proxy schedule $\gamma_t^{\mathrm{proxy}} = \bar{C}_\gamma (t+1)^{-(1+\rho)}$ ($\rho > 0$). Let $\mathcal{F}_t$ denote the σ-field at the start of round t and $\mathbb{E}_t[\cdot] \triangleq \mathbb{E}[\cdot \mid \mathcal{F}_t]$.

Server update decomposition.

$$\omega_{t+1} = \sum_{k\in S_t} \alpha_{k,t}\,\omega_{t+1}^k + \beta_t\,\omega_t^{\mathrm{proxy}} \tag{2}$$

$$= \omega_t - \eta_t E\,(g_t + b_t) + r_t. \tag{3}$$

This identity separates the update into three components: $-\eta_t E g_t$ (ideal gradient step), $-\eta_t E b_t$ (bias from partial participation and dynamic weights), and r_t (proxy residual decaying with t). Define the unified constant

$$c_g \triangleq 4\max\big\{\, 1 + 8L^2\bar{\eta}^2E^2,\ M(1+8L^2\bar{\eta}^2E^2)\,\big\}, \tag{4}$$

where $\bar{\eta}$ bounds η_t and $c_g \geq 4M$.

A. Assumptions

Assumption 1 (Smoothness, Unbiasedness, Variance). *Each F_k is L-smooth with $F_{\inf} \triangleq \inf_\omega F(\omega) > -\infty$; $\mathbb{E}[g_{t,e}^k \mid \mathcal{F}_t, \omega_{t,e}^k] = \nabla F_k(\omega_{t,e}^k)$; $\mathbb{E}_t\|g_{t,e}^k - \nabla F_k\|^2 \leq \sigma_l^2$.*

Assumption 2 (Heterogeneity and Boundedness). *$\exists M \geq 1$, $\kappa \geq 0$ such that $\sum_k p_k\|\nabla F_k(\omega)\|^2 \leq \kappa^2 + M\|\nabla F(\omega)\|^2$. Adaptive coefficients are bounded ($\lambda_t^k \leq \lambda_{\max}$, $\mu_t \leq \mu_{\max}$), with states in a compact set. Every round $S_t \neq \varnothing$ and $\pi_k \geq \pi_{\min} > 0$.*

Assumption 3 (Algorithmic Bridging). *$\|\bar{g}_t^k\| \leq G$ almost surely; $\sup_t \|\omega_t^{\mathrm{proxy}} - \omega_t\| \leq R_\beta$.*

Proposition 1 (Compactness and Projection). *If a compact set $\mathcal{K}_\omega$ contains all $\omega_t, \omega_{t,e}^k, \omega_t^{\mathrm{proxy}}$ and the stochastic gradient map is continuous, then Assumption 3 holds (with G equal to the maximum gradient norm on the compact set and $R_\beta = \mathrm{diam}(\mathcal{K}_\omega)$). With projection-clipping and Euclidean projection, boundedness conditions in Assumption 2 also hold.*

B. Bias and Residual

Proposition 2 (Bias Decomposition). *$\|b_t\| \leq 2G$. Decompose $b_t = m_t + \zeta_t$ with $m_t \triangleq \mathbb{E}_t[b_t]$ and $\mathbb{E}_t[\zeta_t] = 0$. By Jensen's inequality, $\|m_t\|^2 \leq 4G^2$ and $\mathbb{E}_t\|\zeta_t\|^2 \leq 4G^2$. Denote $B_\mu^2 \triangleq 4G^2$ and $B_V^2 \triangleq 4G^2$ (constant bounds). Using only this constant bound, convergence is to an $\mathcal{O}(G^2)$ error neighborhood rather than exact stationarity.*

Proposition 3 (Proxy Residual). *$\|r_t\|^2 \leq D/(t+1)^{2+2\rho}$ with $D \triangleq (\pi_{\mathrm{proxy}}\bar{C}_\gamma R_\beta/\pi_{\min})^2$; hence $\sum_{t=0}^\infty \big(\frac{8}{\eta_t E} + \frac{3L}{2}\big)\,\mathbb{E}\|r_t\|^2 < \infty$.*

C. Core Lemmas

Lemma 1 (Client Drift Bound). *Define $D_e \triangleq \sum_k p_k\,\mathbb{E}_t\|\omega_{t,e}^k - \omega_t\|^2$. If $8L^2\eta_t^2E^2 \leq 1$, then for all $e \leq E$,*

$$D_e \leq 2\eta_t^2 e^2\big(2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F(\omega_t)\|^2\big). \tag{5}$$

Proof sketch. From $\omega_{t,e}^k - \omega_t = -\eta_t\sum_{i=0}^{e-1} g_{t,i}^k$, Cauchy–Schwarz yields $D_e \leq \eta_t^2 e\sum_{i=0}^{e-1}\sum_k p_k\,\mathbb{E}_t\|g_{t,i}^k\|^2$. Decompose the stochastic gradient as $g_{t,i}^k = \nabla F_k(\omega_{t,i}^k) + \varepsilon_{t,i}^k$. By Assumption 1 and $\|a+b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$: $\mathbb{E}_t\|g_{t,i}^k\|^2 \leq 2\sigma_l^2 + 2\|\nabla F_k(\omega_{t,i}^k)\|^2$. Pull $\nabla F_k(\omega_{t,i}^k)$ back to ω_t via L-smoothness: $\|\nabla F_k(\omega_{t,i}^k)\|^2 \leq 2\|\nabla F_k(\omega_t)\|^2 + 2L^2\|\omega_{t,i}^k - \omega_t\|^2$. Combining, weighting by p_k, and applying the heterogeneity assumption 2 gives $\sum_k p_k\,\mathbb{E}_t\|g_{t,i}^k\|^2 \leq A_t + 4L^2D_i$ with $A_t \triangleq 2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F\|^2$. Substituting back, defining $D_{\max,e}$, and using the condition $8L^2\eta_t^2E^2 \leq 1$ yields $D_{\max,e} \leq 2\eta_t^2e^2A_t$. The full five-step derivation appears in Appendix A. □

Lemma 2 (Gradient Bias and Second Moment). *Define $\delta_t \triangleq \mathbb{E}_t[g_t] - \nabla F(\omega_t)$. Under the same conditions:*

$$\mathbb{E}_t\|\delta_t\|^2 \leq 2L^2\eta_t^2E^2A_t, \tag{6}$$

$$\mathbb{E}_t\|g_t\|^2 \leq c_g\,(\sigma_l^2 + \kappa^2 + \|\nabla F(\omega_t)\|^2). \tag{7}$$

Proof sketch. $\delta_t = \frac{1}{E}\sum_{e,k} p_k(\nabla F_k(\omega^k_{t,e}) - \nabla F_k(\omega_t))$. By Jensen's inequality and L-smoothness: $\mathbb{E}_t\|\delta_t\|^2 \le \frac{L^2}{E}\sum_e D_e \le L^2 D_{\max,E} \le 2L^2\eta_t^2E^2A_t$. For the second moment: $\mathbb{E}_t\|g_t\|^2 \le A_t + 4L^2D_{\max,E} \le (1+8L^2\eta_t^2E^2)A_t$. Substituting $A_t = 2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F\|^2$ and bounding termwise via c_g (Eq. (4)) yields Eq. (7). See Appendix A for details. □

D. Single-Step Descent and Convergence

Lemma 3 (Single-Step Descent)**.** *If the step-size condition $x_t \triangleq \eta_t EL \le c_0 \le \frac{1}{16(1+c_g)}$ holds, then*

$$\begin{aligned}\mathbb{E}_t[F(\omega_{t+1})] \le F(\omega_t) - \frac{\eta_t E}{8}\|\nabla F(\omega_t)\|^2 \\ + c_1\eta_t^2E^2(\sigma_l^2+\kappa^2+B_V^2) + c_2\eta_t E\cdot B_\mu^2 \\ + \left(\frac{8}{\eta_t E}+\frac{3L}{2}\right)\mathbb{E}_t\|r_t\|^2,\end{aligned} \tag{8}$$

with $c_1 \triangleq 8Lc_0 + \frac{3L}{2}c_g + 3L$ and $c_2 \triangleq 2 + 3c_0$.

Proof sketch. From L-smoothness, expand $F(\omega_{t+1})$ around ω_t with $\Delta_t = -\eta_t E(g_t + b_t) + r_t$. The proof proceeds in three steps: **Step 1:** Expand $\mathbb{E}_t\langle\nabla F, \Delta_t\rangle$ into three inner-product terms (A)–(C) using $\mathbb{E}_t[g_t] = \nabla F + \delta_t$; apply Young's inequality to each and sum. **Step 2:** Bound the quadratic term $\|\Delta_t\|^2$ by separating g_t, b_t, r_t and substituting the second-moment bounds from Lemma 2. **Step 3:** Merge the inner product and quadratic bounds, substitute the bound on $\|\delta_t\|^2$ from Eq. (6), absorb positive $\|\nabla F\|^2$ terms via the step-size condition, and combine the remaining terms to obtain c_1 and c_2. The full five-step algebraic derivation appears in Appendix A. □

Theorem 1 (Surrogate Objective Convergence)**.** *Let $\eta_t = \eta_0/\sqrt{t+1}$ and assume the step-size condition holds. Under Assumptions 1, 2, and 3:*

$$\begin{aligned}\min_{0\le t<T}\mathbb{E}\|\nabla F(\omega_t)\|^2 \le \frac{8\Delta_F}{E\eta_0\sqrt{T}} + \frac{8\mathcal{R}_\infty}{E\eta_0\sqrt{T}} + 32c_2G^2E \\ + \frac{8c_1E\eta_0(1+\ln T)}{\sqrt{T}}(\sigma_l^2+\kappa^2+4G^2),\end{aligned} \tag{9}$$

where $\Delta_F \triangleq F(\omega_0) - F_{\inf}$ and $\mathcal{R}_\infty \triangleq \sum_{t=0}^\infty(\frac{8}{\eta_t E} + \frac{3L}{2})\,\mathbb{E}\|r_t\|^2 < \infty$.

Proof sketch. Take full expectation of Eq. (8) and telescope-sum over $t = 0$ to $T-1$ ($S_T \triangleq \sum_{t=0}^{T-1}\eta_t$). Using $F(\omega_T) \ge F_{\inf}$, rearranging, and extracting the minimum yields $\min_{0\le t<T} \le \frac{8}{ES_T}[\Delta_F + c_1E^2(\cdots)\sum\eta_t^2 + 4c_2G^2E\cdot S_T + \mathcal{R}_\infty]$. With $\eta_t = \eta_0/\sqrt{t+1}$, we have $S_T \ge \eta_0\sqrt{T}$ and $\sum\eta_t^2 \le \eta_0^2(1+\ln T)$. The critical observation is that the systematic bias term becomes $\frac{8}{ES_T}\cdot 4c_2G^2E\cdot S_T = 32c_2G^2E$—the S_T factors cancel, producing a non-decaying **error floor**. Full details, including convergence of $\mathcal{R}_\infty$, are in Appendix A. □

Corollary 1 (Convergence to Error Neighborhood)**.** $\limsup_{T\to\infty}\min_{0\le t<T}\mathbb{E}\|\nabla F(\omega_t)\|^2 = 32c_2G^2E.$

E. Extension to the Adaptive Objective

Assumption 4 (Regularity and Drift)**.** *$\widetilde{F}_{k,t}$ is L-smooth with the same heterogeneity bound as Assumption 2; $g^k_{t,e}$ is unbiased for $\widetilde{F}_{k,t}$ (variance $\le \sigma_l^2$). There exist ε_t such that $\sup_\omega|\widetilde{F}_{t+1}(\omega) - \widetilde{F}_t(\omega)| \le \varepsilon_t$ and $\sum_{t=0}^\infty\varepsilon_t \le \mathcal{E}_\infty < \infty$.*

Proposition 4 (Sufficient Drift Condition)**.** *Let $z_t = (\{\lambda^k_t\}, \mu_t, \omega_t, \omega^{\text{proxy}}_t, \{W^k_{r,t}\})$ collect the adaptive quantities. If the total variation is finite ($\sum_{t=0}^\infty\|z_{t+1}-z_t\| < \infty$), then the drift condition in Assumption 4 holds with $\varepsilon_t = L_z\|z_{t+1}-z_t\|$, yielding $\mathcal{E}_\infty < \infty$. This is typically satisfied in practice due to decaying adaptive schedules.*

Proposition 5 (Lemma Transfer)**.** *Under the above conditions, Lemmas 1–3 remain valid when $F_k, F, \nabla F$ are replaced by $\widetilde{F}_{k,t}, \widetilde{F}_t, \nabla\widetilde{F}_t$. Key insight: inside $\mathbb{E}_t$, the adaptive parameters are frozen, making $\widetilde{F}_t$ behave as a fixed objective; the proofs depend only on the unified constants L, σ_l, κ, M.*

Theorem 2 (Adaptive Objective Convergence)**.** *With the same η_t and step-size condition, under Assumptions 2, 3, and 4:*

$$\begin{aligned}\min_{0\le t<T}\mathbb{E}\|\nabla\widetilde{F}_t(\omega_t)\|^2 \le \frac{8\widetilde{\Delta}_0}{E\eta_0\sqrt{T}} + \frac{8\mathcal{E}_\infty}{E\eta_0\sqrt{T}} + \frac{8\mathcal{R}_\infty}{E\eta_0\sqrt{T}} \\ + 32c_2G^2E \\ + \frac{8c_1E\eta_0(1+\ln T)}{\sqrt{T}}(\sigma_l^2+\kappa^2+4G^2),\end{aligned} \tag{10}$$

where $\widetilde{\Delta}_0 \triangleq \widetilde{F}_0(\omega_0) - \widetilde{F}_{\inf}$.

Proof sketch. By Proposition 5, Eq. (8) holds for $\widetilde{F}_t$. Using $\widetilde{F}_{t+1}(\omega_{t+1}) \le \widetilde{F}_t(\omega_{t+1}) + \varepsilon_t$ from Assumption 4, telescoping summation introduces $\sum\varepsilon_t \le \mathcal{E}_\infty$, which contributes $\frac{8\mathcal{E}_\infty}{E\eta_0\sqrt{T}}$ after division by S_T—decaying at $\mathcal{O}(1/\sqrt{T})$. The error floor $32c_2G^2E$ remains unchanged. Full details in Appendix A. □

Corollary 2 (Adaptive Error Neighborhood)**.** $\limsup_{T\to\infty}\min_{0\le t<T}\mathbb{E}\|\nabla\widetilde{F}_t(\omega_t)\|^2 = 32c_2G^2E.$

F. Conclusion and Rate Decomposition

Physical origin of the error floor. The error floor $32c_2G^2E$ arises from the non-decaying constant bound $B_\mu^2 = 4G^2$ on the systematic bias m_t in $b_t = m_t + \zeta_t$. During telescoping summation, the $\eta_t E$ contributions from m_t accumulate proportionally to S_T; dividing by S_T cancels the dependence, leaving a nonzero constant. G (Assumption 3) is the maximum client gradient norm, analogous to a fixed noise floor in communication systems, determining the ultimate accuracy ceiling.

Rate decomposition. The upper bound in Theorem 2 consists of five components listed in Table I. As $T\to\infty$, all decaying terms vanish, and the squared gradient norm approaches the neighborhood of $32c_2G^2E$.

TABLE I
CONVERGENCE RATE DECOMPOSITION

Term	Rate
$\frac{8\widetilde{\Delta}_0}{E\eta_0\sqrt{T}}$ (initial potential)	$\mathcal{O}(1/\sqrt{T})$
$\frac{8\mathcal{E}_\infty}{E\eta_0\sqrt{T}}$ (objective drift)	$\mathcal{O}(1/\sqrt{T})$
$\frac{8\mathcal{R}_\infty}{E\eta_0\sqrt{T}}$ (proxy residual)	$\mathcal{O}(1/\sqrt{T})$
$\mathbf{32c_2G^2E}$ **(error floor)**	**non-decaying**
$\frac{8c_1E\eta_0(1+\ln T)}{\sqrt{T}}(\cdots)$	$\mathcal{O}(\ln T/\sqrt{T})$

APPENDIX A
COMPLETE DERIVATIONS

Lemma 1 (Client Drift Bound)

Step 1 — Drift recursion. $\omega_{t,e}^k - \omega_t = -\eta_t \sum_{i=0}^{e-1} g_{t,i}^k$. Cauchy–Schwarz: $\|\omega_{t,e}^k - \omega_t\|^2 \le \eta_t^2 e \sum_{i=0}^{e-1} \|g_{t,i}^k\|^2$. Taking $\mathbb{E}_t$ and weighted summation: $D_e \le \eta_t^2 e \sum_{i=0}^{e-1} \sum_k p_k \,\mathbb{E}_t\|g_{t,i}^k\|^2$.
Step 2a — Noise decomposition. $g_{t,i}^k = \nabla F_k(\omega_{t,i}^k) + \varepsilon_{t,i}^k$ ($\mathbb{E}_t[\varepsilon] = 0$, $\mathbb{E}_t\|\varepsilon\|^2 \le \sigma_l^2$). By $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$: $\mathbb{E}_t\|g_{t,i}^k\|^2 \le 2\sigma_l^2 + 2\|\nabla F_k(\omega_{t,i}^k)\|^2$.
Step 2b — Pullback. $\nabla F_k(\omega_{t,i}^k) = \nabla F_k(\omega_t) + (\nabla F_k(\omega_{t,i}^k) - \nabla F_k(\omega_t))$. L-smoothness: $\|\nabla F_k(\omega_{t,i}^k)\|^2 \le 2\|\nabla F_k(\omega_t)\|^2 + 2L^2\|\omega_{t,i}^k - \omega_t\|^2$.
Step 2c — Combine. $\mathbb{E}_t\|g_{t,i}^k\|^2 \le 2\sigma_l^2 + 4\|\nabla F_k(\omega_t)\|^2 + 4L^2\,\mathbb{E}_t\|\omega_{t,i}^k - \omega_t\|^2$.
Step 2d — Weighted summation. $\sum_k p_k \,\mathbb{E}_t\|g_{t,i}^k\|^2 \le 2\sigma_l^2 + 4\sum_k p_k\|\nabla F_k(\omega_t)\|^2 + 4L^2 D_i$. Using the heterogeneity assumption 2: $\le A_t + 4L^2 D_i$, $A_t \triangleq 2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F\|^2$.
Steps 3–5 — Solve via prefix maximum. Substituting yields $D_e \le \eta_t^2 e^2 A_t + 4L^2\eta_t^2 e \sum_{i=0}^{e-1} D_i$. Defining $D_{\max,e} \triangleq \max_{0\le j\le e} D_j$: $D_{\max,e} \le \eta_t^2 e^2 A_t + 4L^2\eta_t^2 e^2 D_{\max,e}$. Under $8L^2\eta_t^2E^2 \le 1$, we have $4L^2\eta_t^2e^2 \le 1/2$, so $D_{\max,e} \le 2\eta_t^2 e^2 A_t$. □

Lemma 2 (Gradient Bias and Second Moment)

Bias term δ_t: $\delta_t = \frac{1}{E}\sum_{e,k} p_k(\nabla F_k(\omega_{t,e}^k) - \nabla F_k(\omega_t))$. By Jensen's inequality and L-smoothness: $\mathbb{E}_t\|\delta_t\|^2 \le \frac{L^2}{E}\sum_{e=0}^{E-1} D_e \le L^2 D_{\max,E} \le 2L^2\eta_t^2E^2A_t$. This establishes Eq. (6).
Second moment $\mathbb{E}_t\|g_t\|^2$: $\mathbb{E}_t\|g_t\|^2 \le A_t + 4L^2D_{\max,E} \le (1+8L^2\eta_t^2E^2)A_t$. Substituting $A_t = 2\sigma_l^2 + 4\kappa^2 + 4M\|\nabla F\|^2$:

$$(1+8L^2\eta_t^2E^2)A_t = 2(1+8L^2\eta_t^2E^2)\sigma_l^2 + 4(1+8L^2\eta_t^2E^2)\kappa^2 + 4M(1+8L^2\eta_t^2E^2)\,\|\nabla F\|^2.$$

Using $\eta_t \le \bar{\eta}$ and the definition of c_g (Eq. (4)): $\frac{1}{2}c_g \ge 2(1+8L^2\bar{\eta}^2E^2)$, $c_g \ge 4(1+8L^2\bar{\eta}^2E^2)$, and $c_g \ge 4M(1+8L^2\bar{\eta}^2E^2)$. The three terms are bounded by $\frac{c_g}{2}\sigma_l^2$, $c_g\kappa^2$, and $c_g\|\nabla F\|^2$, summing to $\le c_g(\sigma_l^2+\kappa^2+\|\nabla F\|^2)$. □

Lemma 3 (Single-Step Descent)

Step 1 — Inner product. $\mathbb{E}_t\langle\nabla F, \Delta_t\rangle = -\eta_t E\langle\nabla F, \mathbb{E}_t[g_t]\rangle - \eta_t E\langle\nabla F, m_t\rangle + \mathbb{E}_t\langle\nabla F, r_t\rangle$. Using $\mathbb{E}_t[g_t] = \nabla F + \delta_t$:

$$\text{(A)}\ -\eta_t E\langle\nabla F, \nabla F + \delta_t\rangle = -\eta_t E\|\nabla F\|^2 - \eta_t E\langle\nabla F, \delta_t\rangle$$

$$\text{(B)}\ -\eta_t E\langle\nabla F, m_t\rangle, \qquad \text{(C)}\ \mathbb{E}_t\langle\nabla F, r_t\rangle.$$

Applying Young's inequality (with $\alpha = 1/2$ for (A), $\alpha = 1/2$ for (B), $\alpha = \eta_t E/4$ for (C)):

$$\begin{aligned}\text{(A)} &\le -\eta_t E\|\nabla F\|^2 + \tfrac{\eta_t E}{4}\|\nabla F\|^2 \\ &\quad + \eta_t E\|\delta_t\|^2 = -\tfrac{3\eta_t E}{4}\|\nabla F\|^2 + \eta_t E\|\delta_t\|^2,\\ \text{(B)} &\le \tfrac{\eta_t E}{4}\|\nabla F\|^2 + \eta_t E\|m_t\|^2,\\ \text{(C)} &\le \tfrac{\eta_t E}{8}\|\nabla F\|^2 \\ &\quad + \tfrac{2}{\eta_t E}\,\mathbb{E}_t\|r_t\|^2.\end{aligned}$$

Summing:

$$\begin{aligned}\mathbb{E}_t\langle\nabla F, \Delta_t\rangle &\le -\tfrac{3}{8}\eta_t E\|\nabla F\|^2 \\ &\quad + \eta_t E\|\delta_t\|^2 \\ &\quad + \eta_t E\cdot B_\mu^2 + \tfrac{2}{\eta_t E}\,\mathbb{E}_t\|r_t\|^2.\end{aligned}$$

Step 2 — Quadratic term. $\|\Delta_t\|^2 \le 3\eta_t^2E^2\|g_t\|^2 + 3\eta_t^2E^2\|b_t\|^2 + 3\|r_t\|^2$. Multiplying by $L/2$, taking $\mathbb{E}_t$, and substituting bounds:

$$\begin{aligned}\frac{L}{2}\mathbb{E}_t\|\Delta_t\|^2 &\le \frac{3L}{2}\eta_t^2E^2c_g(\sigma_l^2+\kappa^2+\|\nabla F\|^2)\\ &\quad + \frac{3L}{2}\eta_t^2E^2(B_\mu^2+B_V^2) + \frac{3L}{2}\,\mathbb{E}_t\|r_t\|^2.\end{aligned}$$

Step 3 — Merging. Substituting $\eta_t E\|\delta_t\|^2 \le 8L^2\eta_t^3E^3(\sigma_l^2 + \kappa^2 + M\|\nabla F\|^2)$ (from Eq. (6)):

$$\begin{aligned}\mathbb{E}_t[F(\omega_{t+1})] &\le F(\omega_t) - \tfrac{3}{8}\eta_t E\|\nabla F\|^2 + \tfrac{3L}{2}\eta_t^2E^2c_g\|\nabla F\|^2\\ &\quad + 8L^2\eta_t^3E^3M\|\nabla F\|^2\\ &\quad + \big(\tfrac{3L}{2}c_g\eta_t^2E^2 + 8L^2\eta_t^3E^3\big)(\sigma_l^2+\kappa^2)\\ &\quad + \tfrac{3L}{2}\eta_t^2E^2B_V^2 + \eta_t E B_\mu^2 + \tfrac{3L}{2}\eta_t^2E^2B_\mu^2\\ &\quad + \big(\tfrac{2}{\eta_t E} + \tfrac{3L}{2}\big)\,\mathbb{E}_t\|r_t\|^2.\end{aligned}$$

Step 4 — Coefficient absorption. The coefficient of $\|\nabla F\|^2$ is $-\frac{3}{8}\eta_t E + \frac{3L}{2}\eta_t^2E^2c_g + 8L^2\eta_t^3E^3M = -\frac{3}{8}\eta_t E + \eta_t E\cdot\eta_t EL\cdot(\frac{3}{2}c_g + 8L\eta_t EM)$. Under $\eta_t EL \le c_0 \le \frac{1}{16(1+c_g)}$ and $c_g \ge 4M$, one can verify $\frac{3}{2}c_g + 8L\eta_t EM \le \frac{7}{2}c_g$, so the coefficient satisfies $\le -\frac{3}{8}\eta_t E + \eta_t E\cdot c_0\cdot\frac{7}{2}c_g \le -\frac{\eta_t E}{8}$ (since $\frac{7}{2}c_0c_g < \frac{1}{4}$).
Step 5 — Constants c_1, c_2. Using $8L^2\eta_t^3E^3 \le 8Lc_0\eta_t^2E^2$ to reduce order: $(\frac{3L}{2}c_g + 8Lc_0)\eta_t^2E^2(\sigma_l^2+\kappa^2) + \frac{3L}{2}\eta_t^2E^2B_V^2 \le c_1\eta_t^2E^2(\sigma_l^2+\kappa^2+B_V^2)$, with $c_1 = \frac{3L}{2}c_g + 8Lc_0 + 3L$. The B_μ^2 term: $\eta_t E + \frac{3L}{2}\eta_t^2E^2 = \eta_t E + \frac{3}{2}(\eta_t EL)\eta_t E \le (1+\frac{3}{2}c_0)\eta_t E$, yielding $c_2 = 2 + 3c_0$. □

A. Complete Proofs of Theorems

Theorem 1 (Surrogate Objective Convergence)

Step 1 (Constant setup). From Proposition 2: $\|m_t\|^2 \le B_\mu^2 = 4G^2$, $\mathbb{E}_t\|\zeta_t\|^2 \le B_V^2 = 4G^2$. From Proposition 3: $\|r_t\|^2 \le D/(t+1)^{2+2\rho}$.
Step 2 (Telescoping sum). Take full expectation of Eq. (8) and sum. Let $S_T \triangleq \sum_{t=0}^{T-1}\eta_t$. The five terms accumulate as: leftmargin=*,nosep

- $\mathbb{E}_t[F(\omega_{t+1})] \le \mathbb{E}_t[F(\omega_t)] \implies \mathbb{E}[F(\omega_T)] \le \mathbb{E}[F(\omega_0)]$ (telescoping cancellation)
- $-\dfrac{\eta_t E}{8}\|\nabla F(\omega_t)\|^2 \implies -\dfrac{E}{8}\sum_{t=0}^{T-1}\eta_t\,\mathbb{E}\|\nabla F(\omega_t)\|^2$

- $c_1\eta_t^2E^2(\sigma_l^2+\kappa^2+B_V^2) \implies c_1E^2(\sigma_l^2+\kappa^2+4G^2)\sum_{t=0}^{T-1}\eta_t^2$
- $c_2\eta_tE\cdot 4G^2 \implies 4c_2G^2E\cdot S_T$
- $(\dfrac{8}{\eta_tE}+\dfrac{3L}{2})\mathbb{E}_t\|r_t\|^2 \implies \sum_{t=0}^{T-1}(\dfrac{8}{\eta_tE}+\dfrac{3L}{2})\mathbb{E}\|r_t\|^2$

Step 3 (Rearranging). From $F(\omega_T) \geq F_{\inf}$: $\frac{E}{8}\sum\eta_t\,\mathbb{E}\|\nabla F\|^2 \leq \Delta_F + c_1E^2(\sigma_l^2+\kappa^2+4G^2)\sum\eta_t^2 + 4c_2G^2E\cdot S_T+\mathcal{R}_\infty$.

Steps 4–5 (Extract minimum & series estimates). $\min_{0\leq t<T} \leq \frac{8}{ES_T}[\Delta_F + c_1E^2(\cdots)\sum\eta_t^2 + 4c_2G^2E\cdot S_T + \mathcal{R}_\infty]$. With $\eta_t = \eta_0/\sqrt{t+1}$: $S_T \geq \eta_0\sqrt{T}$ $(T \geq 4)$, $\sum\eta_t^2 \leq \eta_0^2(1+\ln T)$.

Step 6 (Four-term substitution).

$$\frac{8}{ES_T}\Delta_F \leq \frac{8\Delta_F}{E\eta_0\sqrt{T}}$$

$$\frac{8}{ES_T}\,c_1E^2(\cdots)\sum\eta_t^2 \leq \frac{8c_1E\eta_0(1+\ln T)}{\sqrt{T}}(\sigma_l^2+\kappa^2+4G^2)$$

$$\frac{8}{ES_T}\cdot 4c_2G^2E\cdot S_T = 32c_2G^2E \text{ (\textbf{error floor}, } S_T \text{ cancels)}$$

$$\frac{8}{ES_T}\mathcal{R}_\infty \leq \frac{8\mathcal{R}_\infty}{E\eta_0\sqrt{T}}$$

Step 7 (Residual convergence verification). $\mathcal{R}_\infty \leq D\Big[\frac{8}{E\eta_0}\sum(t+1)^{-3/2-2\rho} + \frac{3L}{2}\sum(t+1)^{-2-2\rho}\Big] < \infty$ (since $\rho > 0$). $\square$

Theorem 2 (Adaptive Objective Convergence)

Step 1. Proposition 5 guarantees that Eq. (8) holds for $\widetilde{F}_t$.

Step 2. From $\widetilde{F}_{t+1}(\omega_{t+1}) \leq \widetilde{F}_t(\omega_{t+1}) + \varepsilon_t$ (Assumption 4), taking full expectation:

$$\begin{aligned}\mathbb{E}[\widetilde{F}_{t+1}] \leq \mathbb{E}[\widetilde{F}_t] - \tfrac{\eta_tE}{8}\,\mathbb{E}\|\nabla\widetilde{F}_t\|^2 + c_1\eta_t^2E^2(\cdots)\\ + 4c_2G^2\eta_tE + \big(\tfrac{8}{\eta_tE}+\tfrac{3L}{2}\big)\mathbb{E}\|r_t\|^2 + \varepsilon_t.\end{aligned}$$

Step 3 (Telescoping sum). Summing from $t=0$ to $T-1$: $\frac{E}{8}\sum\eta_t\,\mathbb{E}\|\nabla\widetilde{F}_t\|^2 \leq \widetilde{\Delta}_0 + \sum\varepsilon_t + c_1E^2(\cdots)\sum\eta_t^2 + 4c_2G^2E\cdot S_T+\mathcal{R}_\infty$. Dividing by S_T and taking the minimum: $\frac{8\sum\varepsilon_t}{ES_T} \leq \frac{8\mathcal{E}_\infty}{E\eta_0\sqrt{T}}$; the error floor $32c_2G^2E$ remains unchanged. $\square$